

Computer Music I

MUCP 4341 001

INSTRUCTOR: Ali Balighi
OFFICE: Room 207
OFFICE HOUR: Friday and Monday between 12 and 4 PM (by appointment)
E-MAIL: abalighi@ttu.edu
MEETING TIME: Monday, Wednesday, Friday 11:00 AM – 11:50 AM
SOFTWARE: Audacity, MuseScore, Reaper, etc.

COURSE DESCRIPTION:

Computer Music I provides an introductory survey of digital sound production, synthesis, and composition. Students explore the fundamental concepts and practical applications of computer music across multiple platforms and paradigms. This course serves as a comprehensive introduction to the landscape of computer music technologies, covering audio fundamentals, digital signal processing concepts, and hands-on experience with industry-standard and open-source music software.

COURSE CONTENT:

This course introduces the core concepts and tools of computer music through both theoretical understanding and practical application. Topics include:

- Audio Fundamentals: Sampling, bit depth, sample rate, signal flow, and audio analysis
- Digital Audio Workstations (DAWs): Multitrack recording, MIDI sequencing, audio editing, mixing, and mastering using industry-standard software
- Sound Synthesis: Subtractive, additive, FM, and granular synthesis techniques; oscillators, envelopes, and filters
- Introduction to Real-Time Music Programming: SuperCollider and Pure Data/MaxMSP for algorithmic composition, live coding, and interactive systems
- Hardware Integration: MIDI controllers, audio interfaces, analog synthesizers, and hardware-software integration
- Music Software Ecosystems: Exploring open-source and commercial environments for sound design and composition

COURSE PURPOSE & LEARNING OBJECTIVES:

This course develops foundational competencies in computer music production and sound design. Upon completion, students will be able to:

- Demonstrate understanding of audio fundamentals and digital signal processing concepts
- Operate digital audio workstations and multitrack recording environments effectively
- Design and create sounds using multiple synthesis methods and sound design techniques
- Program music using visual and text-based programming environments (Pure Data, MaxMSP, SuperCollider)
- Compose and produce original music and audio projects incorporating various computer music techniques
- Integrate hardware and software tools for music production and performance
- Critically evaluate different computer music tools and approaches for creative applications

COURSE REQUIREMENTS:

Students are expected to:

- Attend all class sessions and actively participate in discussions and demonstrations
- Complete all assigned homework assignments and technical exercises by due dates
- Bring a laptop with required software installed (links to free options provided)
- Complete reading assignments and online tutorials
- Develop and present a final project demonstrating mastery of course concepts

ASSESSMENT & GRADING:

Final grades reflect student engagement, technical competency, and creative achievement. Evaluation is based on the following components:

Final Project (50%): Original composition/sound design work demonstrating mastery of multiple techniques

Homework Assignments & Technical Exercises (30%): Weekly assignments demonstrating skill development

Class Participation & Engagement (20%): Attendance, active participation, peer collaboration, presentations

Grading Scale: A: 90-100% | B: 80-89% | C: 70-79% | D: 60-69% | F: Below 60%

ATTENDANCE POLICY:

Regular attendance is essential for learning in this hands-on technical course. Each student is allowed up to two excused absences per semester. Additional unexcused absences will result in a one-letter-grade reduction per absence. Excessive absences may result in withdrawal from the course. If you must miss class, notify the instructor by email as soon as possible. Excused absences include documented illness, university business, religious observance, and family emergencies.

COURSE OUTLINE & SCHEDULE:

Fall 2026 - Meets [Monday, Wednesday, Friday at 11 AM]

WEEK 1 (Aug 26-30): Course Overview; Audio Fundamentals (sampling, bit depth, frequency); Sound Demo Tour

WEEK 2 (Sep 2-6): DAW Essentials: Recording, Editing, and Basic Mixing

WEEK 3 (Sep 9-13): MIDI Fundamentals: Note data, controllers, sequencing; HW1 Due (Basic DAW Project)

WEEK 4 (Sep 16-20): Synthesis Foundations: Oscillators, Envelopes, LFOs, Filters

WEEK 5 (Sep 23-27): Subtractive Synthesis & Sound Design; HW2 Due (Synthesizer Patch Design); MIDTERM PRESENTATIONS (Students present their synthesis/sound design projects)

WEEK 6 (Sep 30-Oct 4): Audio Effects Processing: Reverb, Delay, Distortion, EQ, Compression in DAWs

WEEK 7 (Oct 7-11): Basic Mixing & Mastering Techniques; HW3 Due (Mix/Master a Multi-Track Project)

Workshop materials: Fade in/out, add effects, use EQ, export sounds and projects

WEEK 8 (Oct 14-18): Introduction to SuperCollider: Server, Synths, and Real-Time Coding Basics

WEEK 9 (Oct 21-25): SuperCollider: Basic Loops, Sequencing & Algorithmic Composition; HW4 Due (Simple SuperCollider Piece)

WEEK 10 (Oct 28-Nov 1): Introduction to MaxMSP: Visual Programming, Message Boxes, and Interactivity

WEEK 11 (Nov 4-8): MaxMSP Audio Basics: Audio Objects, Synthesis & Simple Effects; HW5 Due (MaxMSP Audio Patch)

WEEK 12 (Nov 11-15): Introduction to Music Notation: MuseScore Studio Basics

Creating scores for different instruments, staves, clefs, time signatures, note entry, articulations, and basic formatting

WEEK 13 (Nov 18-22): Hardware & Software Integration: MIDI Controllers, Audio Interfaces, External Gear Basics

WEEK 14 (Nov 25-29): Thanksgiving Vacation

WEEK 15 (Dec 2-6): Final Project Presentations & Course Wrap-Up

Class schedule is subject to change as per activities and events that unfold during the semester. Final Project proposals due Week 5; Final Project submissions due last day of classes.

FINAL PROJECT GUIDELINES:

Students will create an original computer music composition or interactive sound design project incorporating techniques learned throughout the course. Projects should demonstrate mastery of at least two different technologies/approaches (e.g., DAW + synthesis, Pure Data + hardware, SuperCollider composition, etc.). Projects must be accompanied by documentation including:

- A written description of the project concept and artistic intent (250-500 words)
- Technical documentation explaining software/hardware used and signal flow
- Audio files or video documentation of the completed work
- Code or patch files (where applicable) for reproduction and examination

REQUIRED MATERIALS & SOFTWARE:

All software used in this course is available free or at low cost:

- Audacity (free, open-source audio editor)
- MuseScore Studio (Free, open-source): <https://musescore.org>
- Reaper (free evaluation license) or other DAW of choice
- Access to the course website and online resources

OPTIONAL SOFTWARE:

Digital Audio Workstations (DAWs)

- Cakewalk by BandLab (Free, full-featured): <https://www.bandlab.com/cakewalk>
- GarageBand (Free for macOS/iOS users): <https://www.apple.com/mac/garageband/>

Synthesis & Music Programming

- SuperCollider (Free, open-source real-time audio): <https://supercollider.github.io>
- Pure Data (Free, open-source visual programming): <https://puredata.info>
- Max/MSP Student License (Discounted for students): <https://cycling74.com/max>
- Vital Synth (Free, open-source synthesizer): <https://vital.audio>

MIDI & Audio Tools

- MIDI Monitor (Free MIDI utility): <https://www.snoize.com/MIDIMonitor/>
- Sonic Visualiser (Free audio analysis): <https://www.sonicvisualiser.org>
- VB-Audio Cable (Free virtual audio routing): <https://vb-audio.com/Cable/>

Recording & Editing

- OBS Studio (Free screen/audio recording): <https://obsproject.com>
- OpenShot (Free video editor for audio-visual projects): <https://www.openshot.org>

Music Notation Software

- Dorico SE (Free limited version): <https://www.steinberg.net/dorico/>

TEXAS TECH UNIVERSITY POLICIES

ADA STATEMENT:

Any student who, because of a disability, may require special arrangements to meet the course requirements should contact the instructor as soon as possible to make any necessary arrangements. Students should present appropriate verification from Student Disability Services during the instructor's office hours. Please note: instructors are not allowed to provide classroom accommodation to a student until appropriate verification from Student Disability Services has been provided. For additional information, please contact Student Disability Services in Weeks Hall or call 806-742-2405.

ACADEMIC INTEGRITY STATEMENT:

Academic integrity is taking responsibility for one's own class and/or course work, being individually accountable, and demonstrating intellectual honesty and ethical behavior. Violations of academic integrity include plagiarism, cheating, fabrication, and unauthorized collaboration. All submitted work must be your own unless explicitly stated otherwise by the instructor. Proper citation of sources is required. Violations may result in failing the assignment or course, and referral to the Office of Student Conduct.

RELIGIOUS HOLY DAY STATEMENT:

A student who intends to observe a religious holy day should make that intention known in writing to the instructor prior to the absence. A student who is absent from classes for the observance of a religious holy day shall be allowed to take an examination or complete an assignment scheduled for that day within a reasonable time after the absence.

ACCOMMODATION FOR PREGNANT STUDENTS:

To support the academic success of pregnant and parenting students, the University offers reasonable modifications based on the student's particular needs. Any student who is pregnant or parenting a child up to age 18 may contact Alex Faris, the Texas Tech University designated Pregnancy and Parenting Liaison, at alfaris@ttu.edu or 806.834.3420 to discuss available support.

AI USE POLICY:

Use of generative AI tools (such as ChatGPT) is allowed in this class when properly documented and cited. You may use AI for research, brainstorming, and learning about concepts. However, submitted assignments must represent your own learning and work. Submission of AI-generated content as your own work is a violation of academic integrity. If you use AI tools, clearly document and cite their use in your work.

The instructor reserves the right to modify any part of this syllabus as necessary to meet the learning needs of the class.